



Modelling Financial Markets

Chapter 12: Options and the Volatility Surface



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ ATM: At-The-Money
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CRR: Cox–Ross–Rubinstein (binomial tree)
- ▣ DVOL: Deribit Volatility Index
- ▣ ETF: Exchange-Traded Fund
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ GMM: Generalized Method of Moments
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HAR-RV: Heterogeneous AutoRegressive model of Realised Volatility
- ▣ ITM: In-The-Money
- ▣ MA: Moving Average
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ OLS: Ordinary Least Squares
- ▣ OTM: Out-of-The-Money
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ RV: Realised Variance
- ▣ SSVI: Surface SVI (Gatheral–Jacquier parameterisation of the whole surface)
- ▣ SVI: Stochastic Volatility Inspired (parameterisation)
- ▣ SVIX: Simple VIX: index of simple variance swaps (Martin, 2017)
- ▣ US: United States
- ▣ UTC: Coordinated Universal Time (Temps Universel Coordonné)
- ▣ VIX: CBOE Volatility Index

Acronyms used in this material (2/2)

- VIX3M: Cboe 3-Month Volatility Index
- VIX9D: Cboe 9-Day Volatility Index
- VRP: Variance Risk Premium
- VVIX: Cboe VIX of VIX Index
- XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

Outline

Motivation

Parity and Static Replication

The Binomial Tree

Black–Scholes and the Greeks

Delta Hedging in Practice

Implied Volatility and the Smile

Bitcoin Options and the Volatility Surface

The VIX and the Variance Risk Premium

Options with Zero Days to Expiry

Case Study: Are Option-Implied Bounds Valid and Tight?

AI for Scientific Discovery

Summary

References

Why Options?

What an option does

- An **option** gives the right, not the obligation, to buy or sell an asset at a fixed price
- It pays only in some states of the world: a bet on the **distribution**, not only on the direction
- Option prices reveal what the market expects about volatility and crashes

What we do in this chapter

- Derive prices by replication: the Black–Scholes equation and the static spanning of any pay-off by options
- Derive what a hedge leaves behind: discrete-hedging error and the gamma-weighted volatility P&L
- Estimate the smile and the risk-neutral density under no-arbitrage constraints, with confidence bands
- Test the VIX as a forecast and the variance premium as a predictor, with the right standard errors

Chicago, 1973: Listed Options Begin



Trading floor of the Chicago Board of Trade in 2006; its members founded the Cboe in 1973

Photo: [Richie Diesterheft \(2006\)](#); CC BY 2.0; [Wikimedia](#)

[Commons](#)

- The Chicago Board Options Exchange (Cboe) opened in 1973 as the first exchange for standardised, listed options [MacKenzie & Millo \(2003\)](#)
 - ▶ Before: options were negotiated one by one, over the counter
- The same year: the Black–Scholes and Merton pricing papers [Black & Scholes \(1973\)](#), [Merton \(1973\)](#)
- Traders soon used the formula on the floor: theory changed the market it described [MacKenzie & Millo \(2003\)](#)
- Textbooks: [Franke, Härdle & Hafner \(2019\)](#), [Härdle, Chen & Overbeck \(2017\)](#), [Gatheral \(2006\)](#)

The Formula and Its Authors



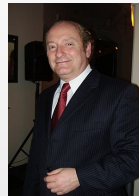
Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia;
public domain; Wikimedia Commons



Myron Scholes (2008)

Photo: Nobellaureatesphotographer (2008);
CC BY-SA 3.0; Wikimedia Commons



Robert C. Merton (2006)

Photo: Digarnick (2006); public domain;
Wikimedia Commons

- Key idea: a portfolio of the stock and cash can **replicate** the option, so its price follows from no-arbitrage [Black & Scholes \(1973\)](#)
- Scholes and Merton received the [1997 Nobel Memorial Prize in Economic Sciences](#); Black had died in 1995

Data Used in This Chapter

Series	What it measures	Period	Owner
S&P 500	US equity index, daily close	1990–2026	S&P Dow Jones Indices
SPY	ETF on the S&P 500: open and close	1993–2026	State Street
VIX	30-day implied volatility of the S&P 500	1990–2026	Cboe
VIX9D, VIX3M	the same over 9 days and 3 months	2011 / 2007–2026	Cboe
VVIX	implied volatility of the VIX	2006–2026	Cboe
Bitcoin options	all listed contracts, one dated snapshot	22 Sep 2026	Deribit
DVOL	30-day implied volatility of Bitcoin	2021–2026	Deribit

- Daily data up to 18 September 2026; the option snapshot was taken on 22 September 2026, 07:51 UTC
- Implied volatility is quoted in annualised %; “vol points” = percentage points of volatility

Notation and Prerequisites

- European call and put pay off $(S_T - K)^+$ and $(K - S_T)^+$ at expiry T ; $\tau = T - t$ in years; American options can be exercised at any time up to T
 - ▶ Both pay-offs are convex in S_T : the value of this convexity is what the smile prices
- Forward $F = Se^{(r-q)\tau}$, log-moneyness $k = \ln(K/F)$; ATM: $k \approx 0$; OTM: puts with $K < F$ and calls with $K > F$; ITM: the opposite
 - ▶ Most of this chapter works in k and in total implied variance $w = \sigma_{imp}^2 \tau$
- No-arbitrage bounds for European options on a non-dividend-paying asset, without any model:
 $\max(S - Ke^{-r\tau}, 0) \leq C \leq S$ and $\max(Ke^{-r\tau} - S, 0) \leq P \leq Ke^{-r\tau}$
 - ▶ So, with $r > 0$ and $\tau > 0$, an American call on a stock without dividends is never exercised early: it is worth at least $S - Ke^{-r\tau} > S - K$; with $r < 0$ early exercise can be optimal
- Assumed: Brownian motion, Itô's lemma and martingales (Chapter 11); regression with HAC standard errors (Chapter 8)

Put–Call Parity

- Two portfolios with the same pay-off at T (European options, same K and T , no dividends):
 - ▶ A: one call + cash Ke^{-rT} (worth K at T) $\Rightarrow \max(S_T, K)$
 - ▶ B: one put + one share $\Rightarrow \max(K, S_T)$
- Same pay-off in every state, so the same price today, otherwise there is an arbitrage:
 - ▶ $C_t - P_t = S_t - Ke^{-rT}$ (with dividend yield q : S_te^{-qT} instead of S_t)
- No model for S is needed: parity uses only no-arbitrage [Merton \(1973\)](#)
- Consequence: a call and a put with the same K and T have the **same** implied volatility (Section 6)

Static Replication: the Carr–Madan Spanning Formula

- Any twice-differentiable pay-off $f(S_T)$ and any point F (Taylor with integral remainder, split at F):

$$\blacktriangleright f(S_T) = f(F) + f'(F)(S_T - F) + \int_0^F f''(K)(K - S_T)^+ dK + \int_F^\infty f''(K)(S_T - K)^+ dK$$

- Proof for $S_T > F$: $\int_F^{S_T} f''(K)(S_T - K) dK = f(S_T) - f(F) - f'(F)(S_T - F)$ by parts; the put integral vanishes; symmetric for $S_T < F$

- Take $e^{-r\tau} E^Q[\cdot]$ with F the forward (a forward contract costs 0):

$$\blacktriangleright V_0 = e^{-r\tau} f(F) + \int_0^F f''(K) P(K) dK + \int_F^\infty f''(K) C(K) dK$$

- A bond, a forward and a strip of OTM options with weights $f''(K)$ replicate any smooth pay-off, whatever the model for S [Bakshi, Kapadia & Madan \(2003\)](#)

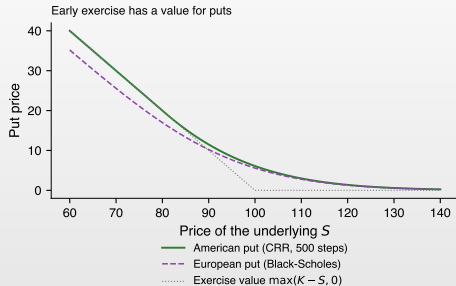
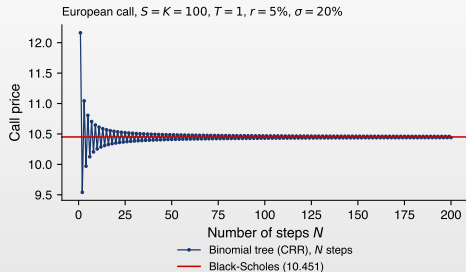
Three Corollaries of Spanning

- ▣ **Variance:** $f(S) = -2\ln(S/F)$ has $f''(K) = 2/K^2$: the strip with weights $1/K^2$ is the variance swap and the VIX (Section 8)
 - ▶ Low strikes get the largest weights, so an expensive put wing makes variance expensive
- ▣ **Density:** a butterfly $[C(K-h) - 2C(K) + C(K+h)]/h^2 \rightarrow \partial^2 C/\partial K^2 = e^{-r\tau} q(K)$ [Breedon & Litzenberger \(1978\)](#): the risk-neutral density is a price (Section 7)
- ▣ **Moments:** $f = [\ln(S/F)]^n$, $n = 2, 3, 4$, price the raw risk-neutral moments of the log return; centring and standardising them gives volatility, skewness and kurtosis [Bakshi, Kapadia & Madan \(2003\)](#); $f = S^2$ gives the SVIX of [Martin \(2017\)](#)
- ▣ Retail strategies (straddle, spread, protective put) are special cases: finite strips with f'' concentrated at one or two strikes

No Arbitrage and the Pricing Measure

- One period, $S \rightarrow uS$ or dS , $d < R = e^{r\Delta t} < u$: $\Delta = \frac{C_u - C_d}{(u - d)S}$ shares plus cash replicate the call, so
 - ▶ $C = \frac{q C_u + (1 - q) C_d}{R}$, $q = \frac{R - d}{u - d}$: q is the unique probability under which S_t/R^t is a martingale
- The real probability of an up move never enters: risk preferences are already in S
- **Fundamental theorem:** no arbitrage $\Leftrightarrow$ an equivalent martingale measure Q exists; complete market $\Leftrightarrow Q$ is unique [Harrison & Pliska \(1981\)](#), [Delbaen & Schachermayer \(1994\)](#)
 - ▶ With jumps or stochastic volatility the market is incomplete: many Q , and option prices carry risk premia (Section 8)
- CRR tree [Cox, Ross & Rubinstein \(1979\)](#): $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, value backwards from T ; American options: $\max(\text{continuation}, \text{exercise})$ at every node

From the Tree to Black-Scholes



□ Left: CRR price of a European call for $N = 1, \dots, 200$ steps; right: American vs European put

Reading the Binomial Results

- The tree converges to Black–Scholes (10.451) as N grows: $N = 50$: 10.411; $N = 200$: 10.441; $N = 1000$: 10.449
 - ▶ Convergence oscillates: $N = 10$ gives 10.253, $N = 11$ gives 10.611 (the strike falls on or between nodes)
- American put at $S = K = 100$: 6.090 vs European 5.574
 - ▶ Early-exercise premium 0.516: deep in the money it pays to receive K now and earn interest
- The tree is the workhorse for American options; Black–Scholes is its limit for European ones
- The error is of order $1/N$ and oscillates between even and odd N : $N \cdot$ error stays near -2.0 and $+1.75$ (Seminar 12, A4)

The Black–Scholes Model

- Price follows a **geometric Brownian motion**: $dS_t = \mu S_t dt + \sigma S_t dW_t$
 - ▶ W_t : Brownian motion, independent Normal increments $W_{t+h} - W_t \sim N(0, h)$
 - ▶ So log returns over τ follow the Normal distribution with variance $\sigma^2 \tau$: σ is the annualised volatility
- Assumptions: constant σ and r , continuous trading, no costs, short selling allowed, no arbitrage
- Continuous-time limit of the tree: the hedge Δ is adjusted at every instant [Black & Scholes \(1973\)](#), [Merton \(1973\)](#)
 - ▶ The drift μ drops out, exactly as the real probability did in the tree
- Chapter 11 derives these processes; here we use their consequences

Derivation: the Black–Scholes Equation

- Replicate $V(S, t)$ with a **self-financing** portfolio: Δ_t shares and cash B_t at r , $\Pi = \Delta S + B$; rebalancing is paid from the portfolio, so $d\Pi = \Delta dS + rB dt$ (no $S d\Delta$ term)
- Itô's lemma: $dV = (V_t + \mu SV_S + \frac{1}{2}\sigma^2 S^2 V_{SS}) dt + \sigma SV_S dW_t$
- Choose $\Delta = V_S$, start from $\Pi = V$ ($B = V - SV_S$): $d(V - \Pi) = (V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} - rB) dt$; both dW_t and μ drop out
- A riskless position that costs nothing must earn nothing: the drift is 0, hence the **Black–Scholes equation**
 - $V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0$, with $V(S, T) = \text{pay-off}$
- That μ is irrelevant is a result, not an assumption; Feynman–Kac: $V = e^{-r\tau} E^Q[\text{pay-off}]$ with S drifting at r under Q

The Black–Scholes Formula

- European call and put, no dividends (with yield q : replace S by $Se^{-q\tau}$):
 - ▶ $C = S\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \quad P = Ke^{-r\tau}\Phi(-d_2) - S\Phi(-d_1)$
 - ▶ $d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau},$ with Φ the standard Normal distribution function
- Reading the two terms:
 - ▶ $\Phi(d_2)$ = risk-neutral probability that the call ends in the money
 - ▶ $\Phi(d_1)$ = number of shares in the replicating portfolio (the delta)
- Five inputs: S, K, τ, r observed; σ is the only one not observed: it must be estimated or implied

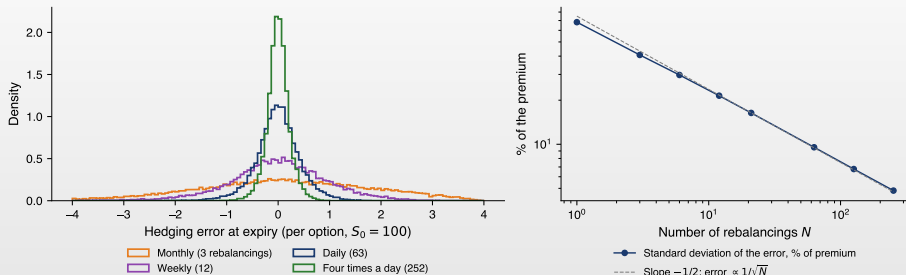
The Greeks: One Identity, All Sensitivities

- Delta $\Delta = \Phi(d_1)$ (put: $\Phi(d_1) - 1$); gamma $\Gamma = \varphi(d_1)/(S\sigma\sqrt{\tau})$, with φ the standard Normal density
- Key identity: $S\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$, because $d_1^2 - d_2^2 = 2\ln(S/K) + 2r\tau$
- Vega: $\partial C/\partial\sigma = S\varphi(d_1)\partial_\sigma d_1 - Ke^{-r\tau}\varphi(d_2)\partial_\sigma d_2 = S\varphi(d_1)\partial_\sigma(d_1 - d_2) = S\varphi(d_1)\sqrt{\tau}$
 - Hence vega = $\Gamma S^2\sigma\tau$: for one expiry, gamma and vega are proportional (Seminar 12, A5–A6)
- Theta from the equation itself: $\Theta = rC - rS\Delta - \frac{1}{2}\sigma^2 S^2\Gamma$; rho = $K\tau e^{-r\tau}\Phi(d_2)$
- Quoting units: vega per vol point, theta per calendar day; call and put with the same K , T share gamma and vega (parity)

Discrete Delta Hedging: the Algorithm

- Sell one call at the Black–Scholes price C_0 with volatility σ_{imp} ; buy Δ_0 shares; the rest is cash
- At each of N rebalancing dates: recompute Δ at the current price and time, buy or sell shares, update the cash with interest
- At expiry: **hedging error** = value of shares and cash minus the option pay-off
- Simulation: geometric Brownian motion, $\mu = 8\%$, $\sigma = 20\%$, three months, 20 000 paths
 - ▶ If the model is right and $\sigma_{imp} = \sigma$, the error vanishes as $N \rightarrow \infty$

Hedging Error and Rebalancing Frequency



□ Left: distribution of the hedging error for four frequencies; right: its standard deviation as % of the premium

 MFM_ch12_hedge_error

Reading the Hedging Error

- The mean error is close to 0 at every frequency: the premium is fair on average
 - ▶ Its standard deviation falls with N : monthly 40.8%, weekly 21.5%, daily 9.5%, four times a day 4.8% of the premium
- Log-log slope -0.48 : the error shrinks like $1/\sqrt{N}$; four times more trades halve it
- Worst 1% of paths: -2.68 with weekly and -1.17 with daily rebalancing, per option on a price of 100
- With transaction costs more trades cost more (a fixed fee per trade grows like N); the best frequency depends on the cost model and on how risk is traded off against cost

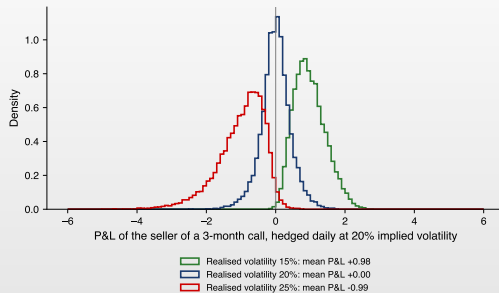
Why $1/\sqrt{N}$? The Discrete Hedging Error

- Over one interval $\Delta t = T/N$ the hedged book earns $\frac{1}{2}\Gamma S^2 [\sigma^2 \Delta t - (\Delta S/S)^2]$: time decay against realised squared return
- Mean 0 under the model to leading order in Δt ; $\text{Var}[(\Delta S/S)^2] \approx 2\sigma^4 \Delta t^2$ and the terms are approximately uncorrelated, so $\text{Var}(\text{error}) \approx \frac{1}{2}\sigma^4 \Delta t E \int_0^T \Gamma_t^2 S_t^4 dt$
- ATM call: $E[\Gamma_t^2 S_t^4] \approx \frac{K^2}{2\pi\sigma^2\sqrt{T^2 - t^2}}$ integrates to $K^2/(4\sigma^2)$, so s.d. $\approx \sigma K \sqrt{T/(8N)} = \sqrt{\pi/4} \mathcal{V}\sigma/\sqrt{N}$
 - With vega $\mathcal{V} = 19.72$ per unit: theory 1.01, 0.44, 0.22 vs simulated 0.96, 0.43, 0.22 for $N = 12, 63, 252$
- Asymptotic theory of the replication error: [Bertsimas, Kogan & Lo \(2000\)](#); it fails with jumps (the error does not vanish as $N \rightarrow \infty$), with stochastic volatility (vega risk) and with costs

Derivation: the Gamma-Weighted Volatility P&L

- ▣ Sell the option at σ_{imp} and hold $\Delta = \partial_S V^{imp}$; the price moves with σ_{real} : $dS = \mu S dt + \sigma_{real} S dW_t$
- ▣ Itô on V^{imp} : $dV = V_t dt + \Delta dS + \frac{1}{2} \Gamma \sigma_{real}^2 S^2 dt$; the equation at σ_{imp} gives $V_t = rV - rS\Delta - \frac{1}{2} \sigma_{imp}^2 S^2 \Gamma$
- ▣ Short option + Δ shares + cash at r , self-financing: $d \text{P\&L} = r \text{P\&L} dt + \frac{1}{2} \Gamma_t S_t^2 (\sigma_{imp}^2 - \sigma_{real}^2) dt$, so over the life
 - ▶ $\text{P\&L} = \int_0^T e^{r(T-t)} \frac{1}{2} \Gamma_t S_t^2 (\sigma_{imp}^2 - \sigma_{real,t}^2) dt$: path-dependent, weighted by Γ , large near K and near T
- ▣ A short variance swap pays $\sigma_{imp}^2 - \sigma_{real}^2$ with constant weights; a delta-hedged option is a variance bet with random weights
- ▣ First-order size: vega $\times$ (implied – realised); here vega = 0.197 per vol point, so 5 vol points $\approx$ 0.99

Selling Volatility at 20%: Three Outcomes

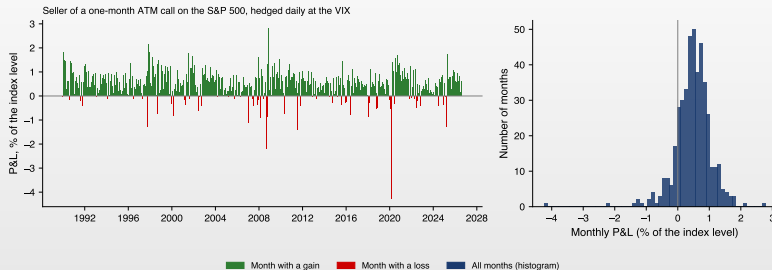


- Three-month call sold at 20% implied volatility, hedged daily; realised volatility 15%, 20% or 25%

Reading the Volatility Bet

- ▣ Realised 15%: mean P&L $+0.98$, positive on 99.8% of paths
- ▣ Realised 20%: mean $+0.00$: the premium was fair, only the discrete-hedging noise remains (s.d. 0.43)
- ▣ Realised 25%: mean -0.99 , positive on only 2.2% of paths
- ▣ Each result is close to $\text{vega} \times 5 \text{ vol points } (\pm 0.99)$: a delta-hedged option is a clean bet on volatility
- ▣ Next: which of the three worlds did the S&P 500 live in?

Thirty-Six Years of Selling S&P 500 Volatility



- Every 21 trading days: sell a one-month at-the-money call priced with the VIX, hedge it daily at the same volatility; P&L in % of the index; interest and dividends ignored

 MFM_ch12_delta_hedged_sp500

Reading the Delta-Hedged History

- ▣ 440 months from 1990-01: mean P&L 0.48% of the index per month, Newey–West $t = 16.0$ [Newey & West \(1987\)](#)
 - ▶ Positive in 86% of months; mean premium collected 2.22%
- ▣ The VIX exceeded the realised volatility of the month in 84% of months (means 19.3 vs 15.3)
- ▣ Worst month -4.25% (2020-03); skewness -1.46 : many small gains, a few large losses
- ▣ Stylised back-test: premiums come from the VIX, an option-strip index, not from quotes of the ATM call; the result is consistent with a **volatility risk premium**, but actual seller returns need historical option quotes and trading costs [Bakshi & Kapadia \(2003\)](#)

Recap: Pricing and Hedging

- Replication gives the Black–Scholes equation; the price depends on one unobserved number, σ
- Discrete hedging leaves an error $\approx \sqrt{\pi/4} \mathcal{V}\sigma/\sqrt{N}$; a wrong σ leaves the gamma-weighted variance P&L
- In a stylised VIX-priced back-test on the S&P 500, sellers of hedged options were paid on average, with crash risk: which volatility does the market use?

Implied Volatility

- **Implied volatility** σ_{imp} : the σ that makes the Black–Scholes price equal the market price
 - ▶ $BS(S, K, \tau, r, \sigma_{imp}) = C^{market}$; unique because vega > 0 : the price increases in σ
- Newton's method: $\sigma_{k+1} = \sigma_k - \frac{BS(\sigma_k) - C^{market}}{\mathcal{V}(\sigma_k)}$
 - ▶ Example: $S = K = 100$, three months, $r = 2\%$, market price 4.50; start at $\sigma_0 = 20.00\%$
 - ▶ $BS(\sigma_0) = 4.2322$, $\mathcal{V} = 19.848 \Rightarrow \sigma_1 = 21.3495\%$; the next step changes only the 6th decimal: $\sigma_{imp} = 21.3495\%$
- Traders quote options in implied volatility, not in money: it is comparable across strikes, maturities and assets
- If Black–Scholes were right, all options on one asset would have the **same** implied volatility

After October 1987: the Smile Appears



Amsterdam stock exchange after the crash, 21 October 1987

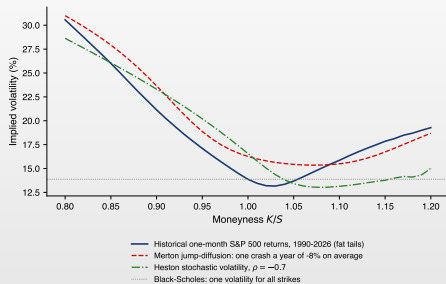
Photo: [Bart Molendijk / Anefo \(1987\)](#); CC0; [Wikimedia Commons](#)

- Before 1987 index options had a nearly flat implied volatility across strikes [Rubinstein \(1994\)](#)
 - ▶ Since the crash, low-strike puts are much more expensive: the skew
- Option prices showed crash fears from October 1986 to August 1987, but not strongly in the two months immediately before the crash [Bates \(1991\)](#)
- **Smile**: implied volatility as a function of K ; **term structure**: as a function of T ; **surface**: both
- Crash risk on the Bucharest Stock Exchange: [Pele & Mazurencu-Marinescu \(2012\)](#)

Why Do Smiles Exist?

- **Fat tails:** large moves are more frequent than the Normal distribution allows (Chapter 1)
 - ▶ Deep OTM options pay only after large moves, so they are worth more than Black–Scholes says
- **Leverage and volatility feedback:** when prices fall, volatility rises [Christie \(1982\)](#)
 - ▶ Negative skewness of returns: the left tail is longer, so low-strike puts are dearer
- **Crash fear and demand for insurance:** investors pay a premium for puts [Bates \(1991\)](#)
 - ▶ Intermediaries who sell them need compensation for crash risk they cannot hedge
- Models: jumps [Merton \(1976\)](#), stochastic volatility [Heston \(1993\)](#), [Hull & White \(1987\)](#), local volatility [Gatheral \(2006\)](#)

Three Sources of a Smile

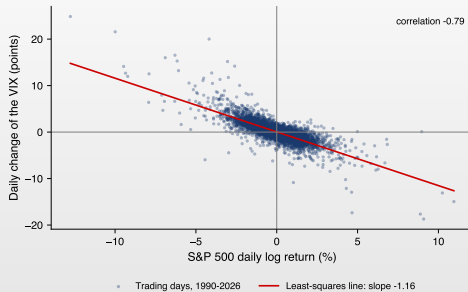


- One-month implied volatility from: the historical distribution of S&P 500 monthly returns (rescaled to mean S , a martingale with $r = 0$, no risk adjustment), a jump model and a stochastic-volatility model

Reading the Model Smiles

- ▣ Historical S&P 500: 9 225 overlapping one-month returns, skewness -1.27 , excess kurtosis 6.8
 - ▶ Implied volatility 30.6% at $K/S = 0.8$, 13.9% at the money, 19.3% at 1.2 : fat tails and negative skewness together create a skewed smile
- ▣ Merton jumps (one crash of about -8% a year): 31.0% vs 16.3% : jumps steepen short-dated smiles
- ▣ Heston with $\rho = -0.7$: 28.6% vs 16.6% vs 13.1% : negative correlation between price and variance tilts the smile into a skew
- ▣ This benchmark has no risk adjustment; risk-neutral index distributions are more left-skewed than realised ones [Jackwerth \(2000\)](#), so priced crash risk steepens market smiles further

The Leverage Effect in the Data



- Daily S&P 500 log return vs the change of the VIX on the same day, 1990–2026

Reading the Leverage Effect

- ▣ Correlation -0.79 over 9 244 days: implied volatility rises when the market falls
- ▣ Slope -1.16 VIX points per 1% return; steeper on down days (-1.39) than on up days (-1.01)
 - ▶ Mean VIX change: $+0.88$ on down days, -0.76 on up days
- ▣ This is the $\rho < 0$ of the Heston model and the asymmetry of GJR-GARCH (Chapter 5)
- ▣ A long position in low-strike puts gains from both the fall and the rise in volatility: this double hedge is why they are expensive

Models of the Smile

Model	Idea	Fits best
Jumps Merton (1976)	Poisson jumps added to the Brownian motion	steep short-dated smiles
Stochastic volatility Heston (1993)	variance follows its own mean-reverting process, correlated with S	skew and term structure of longer maturities
Local volatility Gatheral (2006)	$\sigma(S, t)$ read off the whole surface (Dupire's formula)	matches every quoted price today, exotic options
Parametric smile, SVI Gatheral & Jacquier (2014)	five parameters per expiry, no dynamics	interpolation and quoting (next section)

- Chapter 11 introduces the jump and stochastic-volatility processes; [Franke, Härdle & Hafner \(2019\)](#) and [Čížek, Härdle & Weron \(2011\)](#) give the implied-volatility surface in detail

Recap: Implied Volatility

- ▣ Implied volatility is the market price of an option translated into one number
- ▣ It varies with the strike (smile, skew) and the maturity (term structure): Black–Scholes is a quoting convention, not a model of the data
- ▣ Fat tails, leverage and crash fear all tilt the smile towards low strikes
- ▣ Next: a full surface from a real market, Bitcoin options

Bitcoin Options on Deribit

- Deribit lists European options on Bitcoin and Ether, traded 24/7
 - ▶ **Inverse** options: premium and pay-off are paid in Bitcoin; the price in USD is the BTC premium times the Bitcoin price
- Daily, weekly, monthly and quarterly expiries; each expiry has its own forward price F
- DVOL: Deribit's 30-day implied-volatility index for Bitcoin [Deribit DVOL](#)
- Our snapshot: 22 September 2026, 07:51 UTC; 9 expiries with at least two days left, 375 quoted OTM options
 - ▶ Cryptos as an asset class and their risk: [Pele et al. \(2023\)](#); hedging Bitcoin options: [Matic, Packham & Härdle \(2023\)](#)

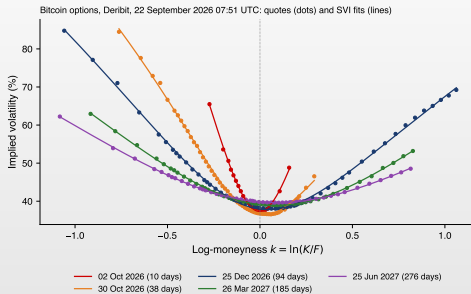
A Parametric Smile: SVI

- Work with the **total implied variance** $w(k) = \sigma_{imp}^2(k) \tau$ at log-moneyness $k = \ln(K/F)$
 - ▶ SVI (stochastic volatility inspired) [Gatheral & Jacquier \(2014\)](#): $w(k) = a + b \left(\rho(k - m) + \sqrt{(k - m)^2 + s^2} \right)$
- Parameters: a level, b slope of the wings, ρ tilt (negative = put skew), m shift, s curvature at the bottom
 - ▶ Wing slopes $b(1 \pm \rho)$: [Lee \(2004\)](#) bounds the asymptotic slope, $\limsup_{|k| \rightarrow \infty} w(k)/|k| \leq 2$, so $b(1 + |\rho|) \leq 2$ is necessary; $w \geq 0$ needs $a + bs\sqrt{1 - \rho^2} \geq 0$
- No arbitrage: call prices must fall and be convex in K (no butterfly arbitrage) and w must increase with τ (no calendar arbitrage)
 - ▶ One SVI fit per expiry by least squares; the exact conditions are on the next slide

No-Arbitrage Constraints on the Smile

- **Wings**, moment formula of [Lee \(2004\)](#): $\beta_R = \limsup_{k \rightarrow \infty} w(k)/k \in [0, 2]$ and $\tilde{p} = \frac{1}{2\beta_R} + \frac{\beta_R}{8} - \frac{1}{2}$
 - ▶ $\tilde{p} = \sup\{p : E^Q[S_T^{1+p}] < \infty\}$: a steeper wing means fatter tails; the left wing mirrors it with negative moments
- **Butterfly** (density $q \geq 0$), with $w' = \partial_k w$ [Gatheral & Jacquier \(2014\)](#):
 - ▶ $g(k) = \left(1 - \frac{k w'}{2w}\right)^2 - \frac{w'^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2} \geq 0$ for all k
 - ▶ and $d_+(k) = -k/\sqrt{w} + \sqrt{w}/2 \rightarrow -\infty$ as $k \rightarrow \infty$ ([Gatheral & Jacquier \(2014\)](#), Lemma 2.2); for SVI this follows from a right-wing slope below 2
- **Calendar**: $\partial_\tau w(k, \tau) \geq 0$ for every k : total variance cannot fall with maturity
- Slice-by-slice least squares imposes none of them; SSVI, the surface version of SVI, has simple sufficient conditions ([Gatheral & Jacquier \(2014\)](#), Theorems 4.1–4.2)

The Bitcoin Smile



- Mark implied volatilities of OTM options (dots) and SVI fits (lines) for five expiries

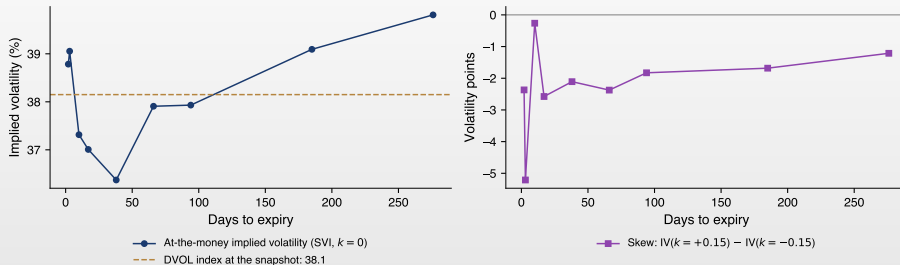
Reading the Bitcoin Smile

- ▣ A true **smile**: both wings rise, unlike the one-sided equity skew
 - ▶ Bitcoin can jump up as well as down: call buyers pay for rallies, put buyers for crashes [Zulfiqar & Gulzar \(2021\)](#)
- ▣ Short expiries have the steepest wings: a jump over a few days matters more relative to $\sigma\sqrt{\tau}$
- ▣ SVI fits the quotes closely: root mean squared error at most 0.76 vol points per expiry
- ▣ At this snapshot puts are slightly dearer than calls: skew from -5.2 to -0.3 vol points

Checking the Nine Bitcoin Fits

- ▣ Wings: largest slopes 0.31 (right) and 0.39 (left), far below 2: Lee's bound holds
 - ▶ Read through Lee: right slope 0.31 $\Rightarrow \tilde{\rho} = 1.18$; a property of SVI's linear extrapolation beyond the last strike, not of the data
- ▣ Butterfly: $\min_k g(k) > 0$ on the grid $k \in [-1.5, 1.5]$ for all 9 expiries (lowest 0.036, 27 November 2026); slopes below 2 give the tail condition
- ▣ Calendar: 1 of 8 adjacent pairs crosses: $w(2 \text{ October } 2026) > w(9 \text{ October } 2026)$ on 24% of $k \in [-0.5, 0.5]$, by at most 0.0075
 - ▶ A calendar arbitrage created by the estimator, not quoted by the market: separate fits ignore the cross-expiry restriction
- ▣ Remedy: estimate the surface jointly under the constraints (SSVI, or least squares with $g \geq 0$ and $\partial_\tau w \geq 0$)

Term Structure and Skew



- Left: ATM implied volatility by expiry and the DVOL index; right: skew = implied volatility at $k = +0.15$ minus at $k = -0.15$

Reading the Term Structure

- ▣ ATM implied volatility between 36.4% and 39.8%; the longest expiry (276 days): 39.8%
 - ▶ DVOL 38.1 vs ATM 37.0% and 36.4% at the expiries around 30 days; a variance-strip index need not lie between ATM volatilities
- ▣ An upward-sloping term structure: the market expects calm now and more uncertainty later
- ▣ The skew is negative for 9 of 9 expiries and closest to 0 at long maturities
- ▣ A snapshot is one day: the level and the sign of the skew change with the market state

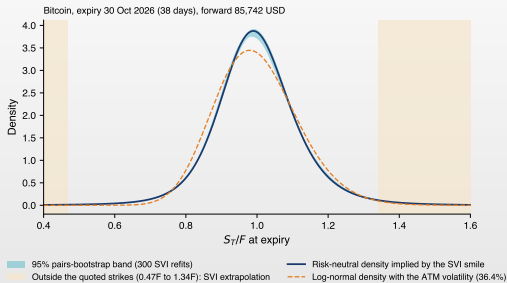
The Risk-Neutral Density

- ▣ Breeden & Litzenberger (1978): the second derivative of call prices in the strike gives the risk-neutral density q of S_T
 - ▶ $q(K) = e^{r\tau} \frac{\partial^2 C(K)}{\partial K^2}$; a butterfly spread $C(K-h) - 2C(K) + C(K+h)$ pays like a narrow bet on $S_T \approx K$
- ▣ Needs a smooth smile: SVI gives $C(K)$ at every strike, then differentiate twice
- ▣ Convexity of C in K (no butterfly arbitrage) is exactly the condition $q \geq 0$
- ▣ q mixes probabilities and risk aversion: comparing it with the physical density gives the **pricing kernel** Aït-Sahalia & Lo (1998), Grith, Härdle & Park (2013)

Estimating the Density: a Statistical Problem

- $q = e^{r\tau} \partial^2 C / \partial K^2$ from noisy, discrete quotes is an ill-posed inverse problem: differentiating twice amplifies the noise
- Estimators: fit a smile (SVI) and differentiate; kernel regression of C on (K, τ) [Aït-Sahalia & Lo \(1998\)](#); the same under monotonicity and convexity in K [Aït-Sahalia & Duarte \(2003\)](#)
 - ▶ Smoothness-penalised implied trees [Jackwerth & Rubinstein \(1996\)](#); Härdle's shape-invariant pricing kernels [Grith, Härdle & Park \(2013\)](#)
- Uncertainty: pairs bootstrap of the quotes, refit, differentiate: pointwise bands for q and intervals for tail probabilities
- Beyond the last quoted strike the data identify nothing: the tail is an extrapolation (SVI wings or an extreme-value tail); report where the quotes end
- Pricing kernel $\propto q/p$ with p the physical density: its non-monotone shape is the pricing-kernel puzzle [Jackwerth \(2000\)](#)

What the Market Prices for Bitcoin



- Risk-neutral density from the SVI smile of the expiry closest to 30 days with a 95% pairs-bootstrap band, vs the log-normal density with the ATM volatility; shaded: outside the quoted strikes

Reading the Risk-Neutral Density

- ▣ Expiry in 38 days, forward 85 742 USD, ATM volatility 36.4%; 56 quotes from $0.47F$ to $1.34F$
 - ▶ Taller centre and fatter tails than the log-normal density: the smile in density form
- ▣ Fall of more than 30%: 1.21%, 95% bootstrap interval $[1.08, 1.24]\%$, vs 0.14% log-normal: about 8 times more
- ▣ Fall of more than 20%: 4.28% $[4.04, 4.46]$ vs 3.27%; rise above 20%: 5.3% vs 5.3%, almost log-normal
- ▣ The band is narrow ($\pm 2.5\%$ at the centre, $\pm 8\%$ at $0.7F$) because the quotes are the exchange's smoothed mark volatilities
 - ▶ It measures fitting noise, not bid-ask uncertainty; $0.7F$ lies inside the quoted strikes, but the probability stays conditional on SVI interpolation and extrapolation: 0.13 of the 1.21 percentage points lie below the lowest quoted strike
- ▣ Prices of insurance, not forecasts: they include the premium for bearing crash risk

Recap: the Volatility Surface

- ▣ A surface is an estimate: one SVI per expiry, checked against Lee's bound, $g(k) \geq 0$ and $\partial_\tau w \geq 0$
- ▣ The risk-neutral density is a second derivative of prices: report bands and where the quotes end
- ▣ Bitcoin: both wings rich, crash probability many times the log-normal one

Variance Swaps

- A **variance swap** pays at T : notional $\times$ (realised variance – strike K_{var}), with realised variance $\frac{252}{n} \sum r_t^2$
 - ▶ The fair strike is the risk-neutral expected variance; with continuous price paths options span it without a model for σ_t (next slides) [Demeterfi, Derman, Kamal & Zou \(1999\)](#)
- Spanning with $f = -2 \ln(S/F)$ (Section 2): a strip of OTM options weighted by $1/K^2$ plus a futures position
 - ▶
$$K_{var} = \frac{2e^{r\tau}}{\tau} \left(\int_0^F \frac{P(K)}{K^2} dK + \int_F^\infty \frac{C(K)}{K^2} dK \right)$$
- Low strikes get the largest weights: expensive puts make variance expensive
- Buyer of variance = insurance against turbulence; seller = earns a premium, pays in crashes [Carr & Wu \(2009\)](#)

Model-Free Implied Variance: What the Strip Measures

- Itô on $\ln S$ (continuous paths): $-2 \ln(S_T/F) = \int_0^T \sigma_t^2 dt - 2 \int_0^T dS_t/S_t$ (with F the forward, $r = q = 0$ for simplicity)
 - ▶ The last term is a Q -martingale, so the strip prices $E^Q[\int_0^T \sigma_t^2 dt]$ for any diffusion [Britten-Jones & Neuberger \(2000\)](#)
- With jumps J : the strip prices $\int \sigma_t^2 dt + 2 \sum (e^J - 1 - J)$, but realised variance is $\int \sigma_t^2 dt + \sum J^2$
 - ▶ Gap $2E^Q \sum (e^J - 1 - J - J^2/2) \approx E^Q \sum J^3/3$: negative for crashes, so the strip understates expected realised variance [Carr & Wu \(2009\)](#)
- Merton model of Section 6 (one jump a year, mean -8%): strip 17.82 vs $\sqrt{E^Q[QV]} = 18.00$ vol points, -2.0% in variance
- “Model-free” means free of a model for σ_t , not of continuity
- SVIX [Martin \(2017\)](#), from $f = S^2$: $SVIX_t^2 = \frac{2}{\tau R_f S_t^2} \left[\int_0^F P dK + \int_F^\infty C dK \right] = \text{var}_t^Q(R_T/R_f)/\tau$, with R_T, R_f gross returns
 - ▶ If $\text{cov}_t(M_T R_T, R_T) \leq 0$ (M_T the stochastic discount factor): $\frac{1}{\tau}(E_t R_T - R_f) \geq R_f SVIX_t^2$, a lower bound on the equity premium

The VIX Index

- Cboe Volatility Index (VIX), since 1993; since 2003 computed from S&P 500 options with the variance-swap formula [Cboe \(2025\)](#)
 - ▶ $\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2$, K_0 = first strike at or below F ; Q = bid-ask mid of the put for $K_i < K_0$, of the call for $K_i > K_0$, the average of both at K_0
 - ▶ $\Delta K_i = (K_{i+1} - K_{i-1})/2$, one-sided at the two ends; options with a zero bid are dropped, and each wing stops after two consecutive zero bids
- Two expiries around 30 days, interpolated to exactly 30 days; $VIX = 100 \sigma$
- VIX futures since 2004, VIX options since 2006 [Cboe \(2025\)](#); the “fear gauge” [Whaley \(2000\)](#)
- Siblings: VIX9D (9 days), VIX3M (3 months), VVIX (the VIX of the VIX)

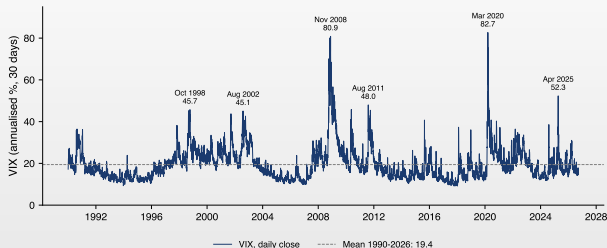
Worked Example: a VIX for Bitcoin

- Cboe rules on our Deribit snapshot: USD mid $Q = \text{BTC mid} \times \text{index } S; F$ and $e^{RT} = F/S$ from Deribit's forward (Cboe: F from parity, R from Treasury yields)
 - ▶ Near expiry: 17 days, 26 OTM options, $\sigma = 39.5\%$
 - ▶ Next expiry: 38 days, 56 OTM options, $\sigma = 40.2\%$
- Interpolate the total variances $\sigma_i^2 T_i$ linearly in time to 30 days:
 - ▶ $\sigma_{30}^2 = \frac{1}{30} \left[T_1 \sigma_1^2 \frac{T_2 - 30}{T_2 - T_1} + T_2 \sigma_2^2 \frac{30 - T_1}{T_2 - T_1} \right]$ (days)
- Our Bitcoin VIX: 40.0; Deribit's DVOL at the same moment: 38.1 [Alexander & Imeraj \(2020\)](#)
- Both exceed the ATM volatility of the two expiries: the variance swap also pays for the wings of the smile

Truncation and Discretisation: How Much Do They Matter?

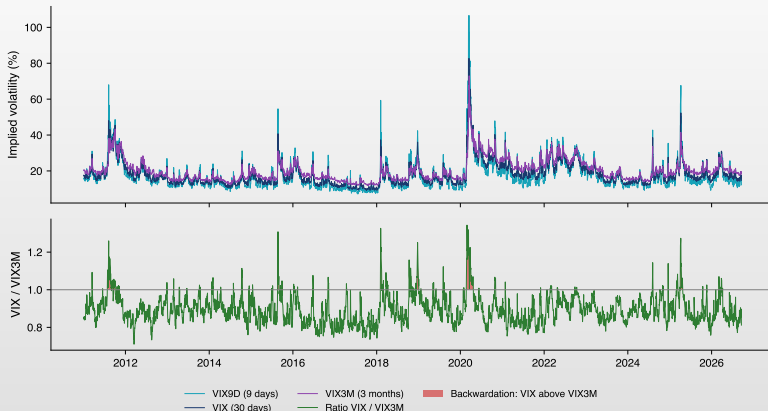
- Only a finite set of strikes is quoted: truncating the strip biases the index down, a coarse grid biases it up [Jiang & Tian \(2005\)](#)
- Bitcoin, same two expiries, 30-day index: quoted strikes 40.03; SVI prices at the same strikes 39.99
 - ▶ SVI on a dense grid inside the quoted range 39.78; extended to $k \in [-4, 4]$ with the SVI wings 40.34
- The near expiry is quoted only from $0.76F$ to $1.17F$: its volatility rises from 39.05 to 40.00 when the wings are added
- These choices move the index by at most 0.6 vol points, less than the gap to DVOL (1.9): the rest reflects the quote selection and methodology of DVOL, not quantified here; an index is an estimator, so report its sensitivity
- A log-normal benchmark with known answer: Seminar 12, A8

The VIX, 1990–2026



- Daily close with the six highest peaks at least 500 days apart
- Mean 19.4 (5.6% a month), range 9.1–82.7 (2020-03-16, COVID-19); above 30 on 7.9% of days: sudden spikes, slow decay, as in GARCH (Chapter 5)

The VIX Term Structure

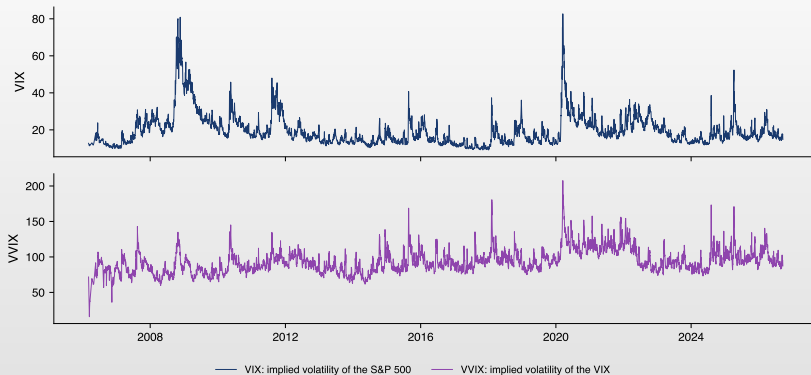


□ Top: implied volatility over 9 days, 30 days and 3 months; bottom: ratio VIX / VIX3M, 2011–2026

Reading the Term Structure

- Normally upward sloping (**contango**): $\text{mean VIX} / \text{VIX3M} = 0.89$; VIX9D below the VIX on 73% of days
 - ▶ In calm markets short-term uncertainty is low and investors pay more for longer protection
- Inverted (**backwardation**) on 7.6% of days, always in crises: fear is concentrated in the near term
- Next 21 days after backwardation: S&P 500 return 2.88% and realised volatility 26.7%, vs 0.80% and 13.6% in contango
- High risk, high expected return after panics: the premium for bearing crash risk [Bollerslev, Tauchen & Zhou \(2009\)](#)

The Volatility of Volatility



□ VIX (top) and VVIX (bottom), daily closes, 2006–2026

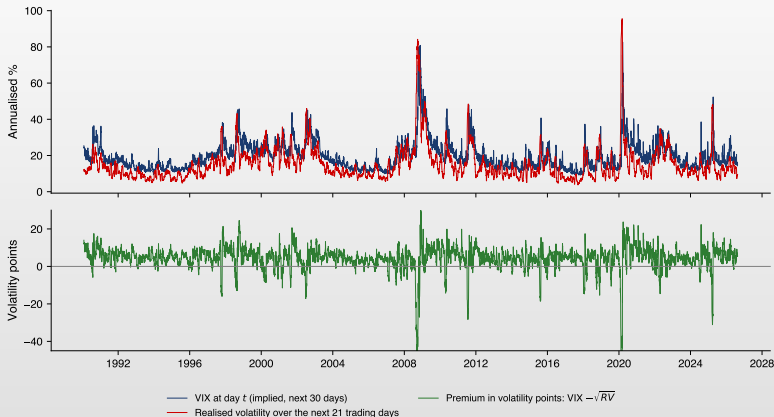
Reading the VVIX

- ▣ Mean VVIX 93.5: the VIX itself is expected to move about 5.3 times more than the index, in relative terms
- ▣ Highest value 207.6 on 2020-03-16, the same day as the VIX peak
- ▣ Daily changes of VIX and VVIX correlate 0.71: when fear rises, uncertainty about fear rises too
- ▣ Volatility is itself random: the core of stochastic-volatility models such as Heston [Heston \(1993\)](#)

The Variance Risk Premium

- Variance risk premium (VRP) Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009); two versions that must not be confused:
 - ▶ **Economic (ex ante):** $E_t^Q[RV_{t,t+21}] - E_t^P[RV_{t,t+21}]$; VIX_t^2 estimates the first term, the second needs a forecast
 - ▶ **Ex post** (used here): $VIX_t^2 - RV_{t,t+21}$, the realised P&L of the seller of a variance swap, known only at $t + 21$
 - ▶ **Ex-ante proxy:** $VIX_t^2 - RV_{t-21,t}$, past variance as the forecast of E_t^P ; known at t , so it can serve as a predictor (last slides of this section); other forecasts give other proxies
 - ▶ $RV_{t,t+21} = \frac{252}{21} \sum_{i=1}^{21} r_{t+i}^2$ (annualised, in %²)
- Positive on average if investors dislike states with high volatility, which are also bad states for wealth
- Overlapping 21-day windows are autocorrelated: Newey–West with at least 21 lags Newey & West (1987); persistent volatility can extend the dependence, so check longer bandwidths

Implied vs Realised Volatility, S&P 500

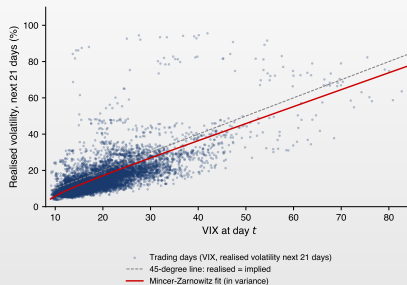


□ Top: VIX and the volatility realised over the next 21 trading days; bottom: their difference, 1990–2026

Reading the Variance Risk Premium

- ▣ 9 203 days: mean VIX 19.4, mean realised volatility 15.3
 - ▶ Premium 4.09 vol points (Newey–West s.e. 0.27); in variance 113 %², $t = 5.5$
- ▣ Positive on 86% of days: implied volatility is usually too high as a forecast
- ▣ The exceptions are the crashes: worst -72.0 vol points, for the window starting 2020-02-20 (COVID-19)
- ▣ Same message as the delta-hedged option: sellers of volatility are paid on average and lose in crises

Is the VIX a Good Forecast?



- Realised volatility over the next 21 days vs the VIX, with the 45-degree line and the Mincer–Zarnowitz fit [Mincer & Zarnowitz \(1969\)](#)

Reading the Forecast Regression

- $RV_{t,t+21} = \alpha + \beta VIX_t^2 + \varepsilon_t$ Mincer & Zarnowitz (1969): an unbiased forecast has $\alpha = 0$, $\beta = 1$; Newey–West errors, 21 lags
 - ▶ Intercept $\hat{\alpha} = -51.9$ (34.4), $\hat{\beta} = 0.86$ (0.10), $R^2 = 0.37$
- Tested separately, neither coefficient differs significantly from its respective null value in $(0, 1)$: $t = -1.51$ and -1.34
 - ▶ But the estimates are correlated -0.90 : the joint Wald test gives $\chi_2^2 = 41.7$, $p = 8.9 \cdot 10^{-10}$: unbiasedness is rejected jointly
- Most points lie below the 45-degree line: the VIX overstates future volatility by the risk premium
- Past realised variance alone: $R^2 = 0.29$; with both, the VIX keeps its weight (0.77) and past variance adds little (0.08, s.e. 0.13)

Is the VIX an Efficient Forecast? Inference Done Right

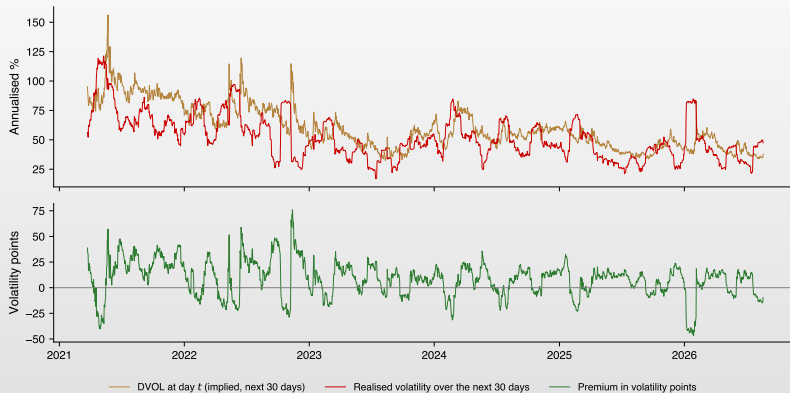
Specification	$\hat{\beta}$	s.e.	$t(\beta = 1)$	Note
Levels, OLS	0.86	0.10	-1.34	joint Wald $p = 8.9 \cdot 10^{-10}$
Logs, OLS	1.06	0.035	1.65	intercept -0.94, $R^2 = 0.57$
Logs, instrumental variables	1.02	0.042	0.60	first-stage $t = 37.7$

- Levels are dominated by 2008 and 2020; in logs the slope is close to 1 and the premium is a proportional shift (intercept < 0)
- VIX^2 measures expected variance with error, which attenuates OLS: instrument it with VIX^2 21 days earlier [Christensen & Prabhala \(1998\)](#); here the slope barely moves
 - ▶ Valid only if lagged VIX^2 is uncorrelated with the forecast error (exclusion, untestable with one instrument): a sensitivity check, not proof
- Newey–West errors with 21 lags; instrumental variables estimated by just-identified GMM with a HAC covariance

Does the VIX Beat HAR-RV? Encompassing and Loss-Based Tests

- Heterogeneous autoregressive model (HAR-RV) [Corsi \(2009\)](#): past daily, weekly and monthly realised variance; encompassing regression in logs [Busch, Christensen & Nielsen \(2011\)](#)
 - ▶ VIX slope 0.90 (0.056); HAR terms jointly $\chi^2_3 = 14.2$, $p = 0.003$; R^2 : VIX 0.570, HAR 0.474, both 0.575
- Out of sample from 2000 (6 697 days; coefficients re-estimated every 21 days on targets already observed)
 - ▶ QLIKE loss $L = RV/\hat{\sigma}^2 - \ln(RV/\hat{\sigma}^2) - 1$, robust to noise in the RV proxy [Patton \(2011\)](#)
 - ▶ Loss ratio VIX^2 / HAR : 0.94, Diebold–Mariano $t = -0.93$ ($p = 0.35$) [Diebold & Mariano \(1995\)](#); MZ-corrected VIX: 0.95 ($p = 0.35$)
- In sample, history adds significant information; out of sample the VIX is about 6% better but not significantly: the evidence is weaker than the in-sample tests suggest
- Targets overlap by 20 of 21 days: the Diebold–Mariano variance is HAC with 21 lags

The Variance Risk Premium of Bitcoin



- DVOL and the Bitcoin volatility realised over the next 30 calendar days, 2021–2026; bottom: their difference

Reading the Bitcoin Premium

- 1975 days: mean DVOL 60.3 vs realised 51.7; premium 8.5 vol points (s.e. 1.6, $t = 5.3$)
 - ▶ Twice the S&P 500 premium in vol points, but similar relative to the level of volatility
- Positive on 71% of days: less often than for the S&P 500; long episodes of negative premium
- Worst: -46.9 vol points for the window starting 2026-01-28
- A young market and a short sample: 66 non-overlapping 30-day periods; a Bitcoin VIX and its premium:
Alexander & Imeraj (2020)

Does the Ex-Ante VRP Predict Returns? Overlapping-Horizon Inference

Horizon	$\hat{b}$	t_{NW}	t_{HH}	$t_{Hodrick}$	p_{boot}	R^2 (%)
$h = 1$	0.07	0.33	0.36	0.41	0.35	0.2
$h = 3$	0.09	0.69	0.67	1.33	0.07	1.2
$h = 6$	0.03	0.49	0.48	0.65	0.26	0.3
$h = 12$	-0.02	-0.64	-0.79	-0.54	0.69	0.2

- Monthly 1990–2026: $\frac{12}{h} \sum_{j=1}^h r_{t+j}^e = a + b \text{VRP}_t + u_t$, $\text{VRP}_t = \text{VIX}_t^2/12 - \text{RV}_t$, ex-ante proxy with past RV_t as the forecast [Bollerslev, Tauchen & Zhou \(2009\)](#)
- If one-month errors are serially uncorrelated, overlap makes u_t an $\text{MA}(h-1)$: Newey–West (h lags), Hansen–Hodrick [Hansen & Hodrick \(1980\)](#), Hodrick 1B [Hodrick \(1992\)](#) (better size); these bandwidths are choices, so check their sensitivity
- p_{boot} : one-sided, VRP simulated as $\text{AR}(1)$ ($\hat{\phi} = 0.13$) under $b = 0$, residuals resampled jointly [Stambaugh \(1999\)](#); low persistence, so little small-sample bias
- No horizon is significant on the full sample; the 1990–2007 evidence is revisited in the last section

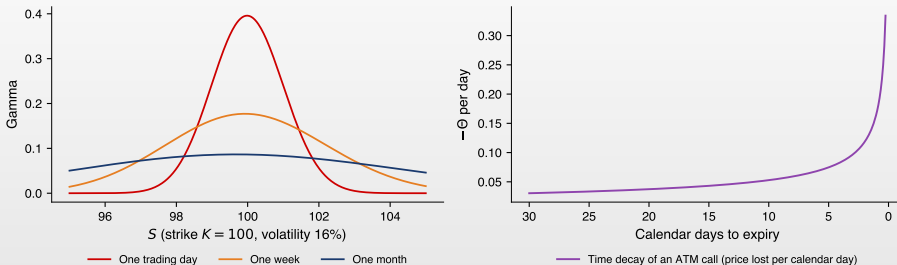
Recap: Implied Volatility as Information

- The VIX is an annualised volatility index: 100 times the square root of an option-strip estimate of the fair strike of a 30-day variance swap on the S&P 500
- Its term structure flags crises (backwardation); VVIX measures uncertainty about volatility
- Implied variance exceeds realised variance on average, for equities and for Bitcoin: the variance risk premium
- As a forecast, the VIX is biased upwards in levels, has slope near 1 in logs, and beats HAR-RV only insignificantly out of sample

Options That Live One Day

- ▣ **0DTE** (zero days to expiry) options: traded on the day they expire [Cboe 0DTE](#)
 - ▶ S&P 500 options now expire every trading day, so a 0DTE contract exists every day
- ▣ More than 75% of retail trades in S&P 500 options involve 0DTE contracts, and retail investors lose money on them [Beckmeyer, Branger & Gayda \(2023\)](#)
 - ▶ 0DTE options were 48% of index-option trading in August 2023 [Brogaard, Han & Won \(2023\)](#)
- ▣ More 0DTE trading is followed by higher index volatility, mostly through speculative retail trades [Brogaard, Han & Won \(2023\)](#); ultra-short pricing needs skewness and kurtosis corrections [Bandi, Fusari & Renò \(2023\)](#)
- ▣ Theory from Section 4: with $\tau \rightarrow 0$ gamma and theta explode at the money

Gamma and Time Decay Near Expiry



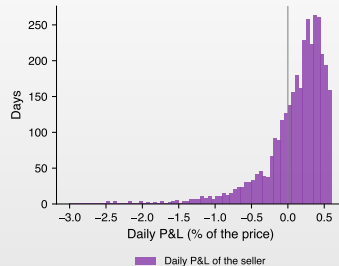
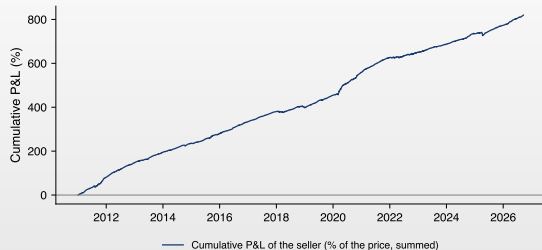
□ Left: gamma of a call with $K = 100$ one day, one week and one month before expiry; right: daily time decay of an ATM call

 MFM_ch12_0dte_gamma

Reading Gamma Near Expiry

- ▣ ATM gamma one day before expiry: 0.396 vs 0.086 one month before: 4.6 times larger ($= \sqrt{21}$)
- ▣ A 1% move in the last day changes delta by about 0.4: the hedge must be adjusted many times within the day
- ▣ Time decay per day: 0.167 in the last day vs 0.030 with 30 days left, about 5.5 times more
- ▣ The seller collects fast time decay and carries large gamma risk: small premium, sudden losses

Selling a Same-Day Straddle: a Stylised Test



- SPY, 2011–2026: sell an ATM straddle at the open, settle at the close; premium = $\sqrt{2/\pi} \sigma_{day} S$ with σ_{day} from the previous VIX9D scaled to the trading session

 MFM_ch12_0dte_straddle

Reading the Straddle Test

- ▣ Session share of daily variance 0.74 (estimated on 1993–2010); 3 950 days
 - ▶ Mean premium 0.76% of the price vs mean absolute open-to-close move 0.55%
- ▣ The seller wins on 75% of days, 20.7 basis points a day on average
- ▣ Worst day -8.25% (2025-04-09), about 11 premia; the move exceeds twice the premium on 3.5% of days
- ▣ Stylised pricing, not traded prices: the result measures the variance premium inside VIX9D
 - ▶ Market 0DTE premia, spreads and intraday hedging would change the numbers [Bandi, Fusari & Renò \(2023\)](#), [Beckmeyer, Branger & Gayda \(2023\)](#)

Case Study: the Research Question

- **Question Back, Crotty & Kazempour (2022):** option prices give lower bounds on the equity premium; are they valid and tight conditionally, and do they forecast returns out of sample?
 - ▶ Why: a tight bound is a real-time equity-premium forecast with no estimated parameter; **Martin (2017)** could not reject $\alpha = 0, \beta = 1$ on 1996–2012, a test with low power
- **Martin bound** (eq. 3), $R_{t,T}$ the gross market return, $R_{f,t,T}$ the gross riskless return, var_t^* the risk-neutral variance (here $\text{SVIX}_{t,T}^2$ is not annualised, unlike Section 8):
 - ▶ $E_t[R_{t,T}] - R_{f,t,T} \geq R_{f,t,T} \text{SVIX}_{t,T}^2$, $\text{SVIX}_{t,T}^2 = \text{var}_t^*(R_{t,T}/R_{f,t,T})$
 - ▶ Valid if $\text{cov}_t(M_{t,T}R_{t,T}, R_{t,T}) \leq 0$ ($M_{t,T}$ the stochastic discount factor); tight if this covariance is zero, e.g. a log-utility investor who holds the market (Section 1)
- **Chabi-Yo–Loudis bound** **Chabi-Yo & Loudis (2020)**: built from the first four risk-neutral moments, free of preference parameters in its restricted form; almost always above the Martin bound (Figure 3.1)
- **Link to this chapter:** SVIX is the static-spanning strip with $f = S^2$ (Sections 2 and 8)
 - ▶ Realised slackness $s_{t,T} = R_{t,T}^e - b_{t,T}$ (excess return minus bound): valid means $E_t[s_{t,T}] \geq 0$, tight means $E_t[s_{t,T}] = 0$

Case Study: Data and Method

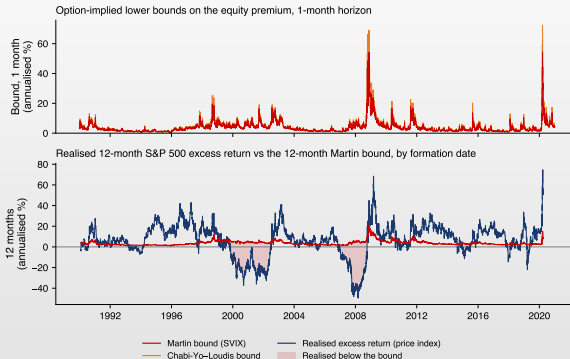
- **Data** (Section 3, Appendix C): daily Martin and Chabi-Yo–Loudis bounds, January 1990 – December 2020, from S&P 500 index options (Cboe data for 1990–1995 and 2020–2021, OptionMetrics for 1996–2019)
 - ▶ Filters: standard options only (no weekly or quarterly), bid > 0, the OTM option at each strike, 7 to 550 days, at least 10 strikes per expiry; bounds interpolated to 30, 90, 180 and 360 days
 - ▶ Returns: S&P 500 with dividends minus the Fama–French risk-free rate, over 21, 63, 126 and 252 trading days up to March 2021, annualised, in %
- **Moments** (eq. 10): for any nonnegative z_t known at t , $E[(R_{t,T}^e - b_{t,T}) z_t] \geq 0$; z_t = a constant and nine Welch–Goyal predictors made positive [Welch & Goyal \(2008\)](#) (Table 3)
- **Kodde–Palm tests** [Kodde & Palm \(1986\)](#) (eqs. 11–14): $\bar{\lambda}$ the 10 sample means, Σ their Hansen–Hodrick covariance [Hansen & Hodrick \(1980\)](#), with lags equal to the days in the horizon
 - ▶ $D_1 = \min_{\lambda \geq 0} (\lambda - \bar{\lambda})^\top \Sigma^{-1} (\lambda - \bar{\lambda})$ tests validity ($H_0: \lambda_0 \geq 0$); $D_2 = \bar{\lambda}^\top \Sigma^{-1} \bar{\lambda} - D_1$ tests tightness ($H_0: \lambda_0 = 0$ vs $\lambda_0 \geq 0$); both follow mixtures of χ^2 laws
- **Forecasting**: bias-corrected predictive regressions [Amihud & Hurvich \(2004\)](#) (Tables 5, 6); monthly out-of-sample forecasts against the expanding post-1926 market mean, 60-month initial window (Table 8)
 - ▶ $R_{OS}^2 = 1 - \sum \varepsilon_t^2 / \sum \nu_t^2$, ε_t and ν_t the errors of the forecast and of the benchmark; Diebold–Mariano tests [Diebold & Mariano \(1995\)](#)

Case Study: Results (Tables 1, 3, 5, 6 and 8)

Horizon	Mean R^e (%)	Mean b (%)	D_1	D_2	$\hat{\beta}$ (s.e.)	R_{OS}^2 (%)
1 month	8.76	3.81	0.00	29.93	1.36 (1.23)	-1.2
3 months	8.73	3.86	0.00	39.51	1.39 (1.47)	-1.2
6 months	8.67	3.83	0.00	49.28	1.78 (0.85)	-0.2
12 months	8.86	3.68	0.00	71.21	1.04 (0.88)	-3.0

- Source: [Back, Crotty & Kazempour \(2022\)](#), *Journal of Financial Economics*, Tables 1, 3, 5, 6 and 8
- **Validity not rejected:** all 10 sample moments are positive, so $D_1 = 0$ and $p = 100\%$ at every horizon (Table 3)
 - ▶ Interpretation: the test based on the nine predictors does not reject validity in 1990–2020; this does not prove validity in every state
- **Not tight:** D_2 exceeds 27.13, the upper critical value of the 0.1% Kodde–Palm test, at every horizon; finite-sample p -values 3.6, 0.4, 2.6, 8.3%, so not rejected at 5% at 12 months; mean slackness 4.8 to 5.2 points a year (Tables 1, 3)
 - ▶ Interpretation: the bound is less than half of the realised premium; validity comes from this average slackness, not from predictive power (Section 4.3)
- **Forecasting:** the slopes $\hat{\beta}$ are positive but imprecise (Table 5); with the nine predictors as controls, a one-standard-deviation rise in the 12-month bound is associated with a 12-month excess return higher by 6.6 points; the bound is significant at 1% at every horizon with asymptotic tests, but bootstrap p -values are 20.7, 0.0, 0.2, 9.5% (Table 6)
 - ▶ Interpretation: the bound carries information, and the significant predictors show that slackness is predictable, so the bound is not tight
- **Out of sample:** the Martin bound as a forecast loses to the historical mean at every horizon; the best case, the Chabi-Yo–Loudis bound at 6 months, reaches 4.3%, not significant (Table 8)
 - ▶ Interpretation: no significant out-of-sample gain is detected in the 30-year sample; low test power is one possible explanation

Case Study: the Bound and the Realised Premium

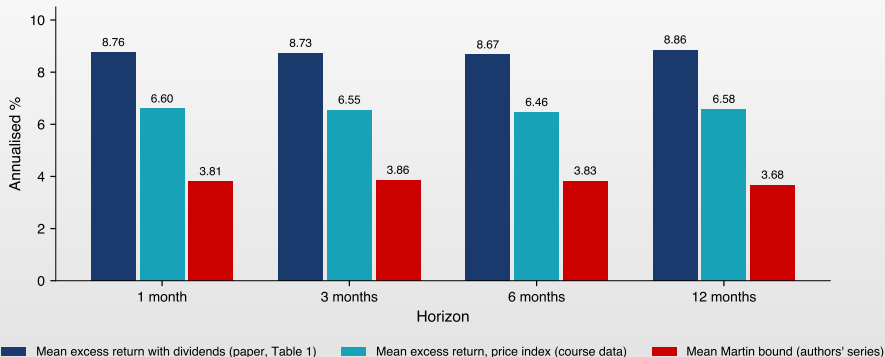


- Top: the authors' daily 1-month bounds, January 1990 – December 2020 ([replication package](#))
- Bottom, our computation on the course data: S&P 500 excess return over the next 12 months (price index minus the Fama–French risk-free rate), formation dates 31 January 1990 – March 2020, against the 12-month Martin bound

Case Study: Reading the Bound Chart

- The 1-month Martin bound has median 2.7% and peaks at 54.4% on 16 March 2020; the 12-month bound peaks at 21.1% on 20 November 2008
 - ▶ The bound is proportional to a risk-neutral variance: it jumps with the VIX in crises, raising the floor for the conditional equity premium
 - ▶ The Chabi-Yo–Loudis bound is at or above the Martin bound on 99.99% of days: the higher moments (negative skewness, fat tails) raise it
- The realised 12-month excess return falls below the 12-month bound on 33% of formation dates, mostly 2000–2002 and 2007–2008
 - ▶ The bound restricts the conditional mean $E_t[R_{t,T}^e]$, not each realisation: validity is tested on averages of $s_{t,T} z_t (D_1)$, not day by day
- Correlation of the 12-month bound with the realised 12-month excess return: 0.15 (overlapping daily windows, descriptive only)
 - ▶ One-year return noise dwarfs the variation of the bound: hence the imprecise slopes of Table 5 and the negative R_{OS}^2 of Table 8

Case Study: Mean Premium and Mean Bound



- Dark blue: Table 1 of the paper (S&P 500 with dividends); teal: our computation on the course data (price index, the same formation days: 7789 at 1 month, 7598 at 12 months); red: mean of the authors' Martin bound, equal to Table 1

Case Study: Reading the Means

- Mean slackness on the course data: 2.79, 2.69, 2.63, 2.91 points a year ($h = 1, 3, 6, 12$), against 4.8 to 5.2 in the paper (Tables 1, 3)
 - ▶ The gap of 2.2–2.3 points is mostly the dividend yield missing from the price index; the bound itself is unchanged
- The mean bound covers 41–44% of the mean premium with dividends and 56–59% of the price-index premium
 - ▶ Mean slackness is positive at every horizon in both cases: validity is not rejected and tightness is easily rejected, as $D_1 = 0$ and the large D_2 of Table 3 show
- Realised excess return below the bound on 41% of days at 1 month and 33% at 12 months: single realisations often violate the bound, the average does not
 - ▶ Seminar C3 runs the Kodde–Palm tests on this smaller slackness

Case Study: Replication, Limitations and an Open Extension

- **Replicate** (seminar C3): the authors' daily bound series ([replication package](#)), S&P 500 closes (data/market), the daily risk-free rate ([Kenneth French Data Library](#)), the Welch–Goyal predictors ([Amit Goyal's data page](#))
 - ▶ Rebuild Table 1 and the statistics D_1 , D_2 of Table 3; the course index has no dividends, so excess returns and slackness are lower: report it as a data limitation
- **Limitations stated by the authors**
 - ▶ A short series, limited to the years after 1990 by the need for option prices: the out-of-sample tests have little power (Section 6)
 - ▶ Validity holds even when the bound is replaced by its time-series mean, so it says nothing about predictive power (Section 4.3)
 - ▶ More recent bounds and formulas for risk premia are not tested (Introduction)
- **Open extension**: is the Martin bound still valid on 2021–2026 (the 2022 bear market, the 0DTE era of Section 9)? The course data have VIX, not SVIX: VIX weights OTM options by $1/K^2$ and exceeds SVIX when the left tail is priced ([Martin \(2017\)](#), Result 3); is a VIX-based bound still valid, and can the Kodde–Palm test detect it if not?

An Open Question: Does the VRP Still Predict Returns?

- [Bollerslev, Tauchen & Zhou \(2009\)](#), Table 2: monthly data 1990–2007, $y_t^{(h)} = \frac{12}{h} \sum_{j=1}^h (r_{m,t+j} - r_{f,t+j})$ on $\text{VRP}_t = \text{VIX}_t^2/12 - \text{RV}_t$
 - ▶ RV_t : realised variance of month t ; at $h = 3$: slope 0.47, Hodrick $t = 2.86$, adjusted $R^2 = 6.82\%$, the peak of a hump-shaped horizon profile
- [Goyal, Welch & Zafirov \(2024\)](#): 29 equity-premium predictors proposed after 2008, re-examined with data to 2021: over one third lose in-sample significance, half of the rest fail out of sample
- Open: is VRP predictability a stable price of variance and jump risk, or a property of 1990–2007? The 2008–2026 sample adds two crashes and the 0DTE era
- Formally: $H_0 : \beta_3 = 0$ vs $H_1 : \beta_3 > 0$ in $y_t^{(3)} = \alpha + \beta_3 \text{VRP}_t + u_t$
 - ▶ Out of sample: H_0 : the VRP forecast does not lower the mean squared prediction error of the historical mean; Clark–West statistic [Clark & West \(2007\)](#)

The Discovery Loop, Step by Step (1/2)

- **Literature:** "List peer-reviewed tests of VRP return predictability since 2009: DOI, sample, horizon, table, headline estimate."
 - ▶ Resolve every DOI on Crossref and read the cited table yourself; search with Semantic Scholar, Google Scholar, Elicit
- **Hypotheses:** "Give three competing explanations of VRP predictability and the testable implication of each for the horizon profile, for 2008-2026 and for non-US markets."
 - ▶ Time-varying risk aversion, jump-risk compensation, or a small-sample artefact of persistent regressors and overlapping returns
- **Data and code:** "Write Python that builds VIX squared/12 and monthly realised variance from data/market/VIX.INDX.csv and GSPC.INDX.csv and reproduces Table 2 of BTZ (2009), $h = 3$, with Hodrick (1992) standard errors."
 - ▶ Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot); replicate the published number before extending anything

The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed before the results: "Draft a pre-analysis plan: horizons, out-of-sample start date, benchmark, test statistic and the multiple-testing correction across horizons."
 - ▶ Placebo: VRP lagged 12 months or block-shuffled; subsamples; 5-minute realised variance of SPY (from 2020) instead of daily squared returns
- **Critique**: "Act as a hostile referee: why could this out-of-sample R^2 be spurious or biased, and which test settles each objection?"
 - ▶ Expect: bias with persistent regressors, overlap, the choice of start date, the weight of 2008 and 2020
- **Report**: every horizon and specification tried, in one table; the plan dated before the results; AI use logged

What the Researcher Must Check

- **Invented or misquoted numbers:** an assistant "recalls" Table 2 of [Bollerslev, Tauchen & Zhou \(2009\)](#)
 - ▶ Test: compare with the paper itself, digit by digit; log any mismatch
- **Timing:** VIX at the end of month t , returns from $t + 1$, T-bill of the same month
 - ▶ Test: recompute the forecast for t on data truncated at t ; it must not change
- **Overlap:** OLS or Newey–West with few lags overstate t for 3-month returns sampled monthly
 - ▶ Test: Hodrick (1992) standard errors [Hodrick \(1992\)](#), or non-overlapping quarterly regressions
- **Horizon and start-date mining:** the best of eight horizons, the most favourable split
 - ▶ Test: report all horizons with a Holm correction; plot R_{OS}^2 against the start date
- **Measurement:** daily squared returns are not 5-minute realised variance
 - ▶ Test: match the published sample means (implied 33.23, realised 14.93) before any regression

Mini-Case: Replicate, Then Extend

- A typical assistant proposal: "the VRP predicts the quarterly equity premium; extend the 2009 result to today". We test it on the course data
- Replication, February 1990 to September 2007, $h = 3$, daily data: $\hat{\beta}_3 = 0.42$, $t_{\text{Hodrick}} = 2.25$, adjusted $R^2 = 4.6\%$ (published: 0.47, 2.86, 6.82%)
 - ▶ Mean implied variance 33.1 is close to 33.23; mean realised 20.6 vs 14.93: daily close-to-close data, a stated limitation
- Extension to May 2026 (436 months): $\hat{\beta}_3 = 0.09$, $t_{\text{Hodrick}} = 1.33$, adjusted $R^2 = 1.0\%$
- Out of sample from January 2008, 221 expanding-window forecasts: $R_{OS}^2 = 1 - \sum_t (y_t - \hat{y}_t)^2 / \sum_t (y_t - \bar{y}_t)^2 = -22.2\%$ [Campbell & Thompson \(2008\)](#); Clark–West $t = 0.64$ ($p = 0.26$): H_0 not rejected
- Reading: sign and size replicate in-sample; the longer sample does not support stable predictability. Outliers, measurement or a vanished premium: that is the open part

 [ai_discovery_case.py](#)

Your Turn: a Project Seed

- Question: does the VRP predict the equity premium after 2008, and does the Bitcoin premium of Section 7 (DVOL) predict Bitcoin returns?
- Replicate first: [Bollerslev, Tauchen & Zhou \(2009\)](#), Table 2, $h = 3$: slope 0.47, $t = 2.86$, adjusted $R^2 = 6.82\%$; explain every gap
- Deliverables
 - ▶ A dated pre-analysis plan; the replication table; out-of-sample results for all horizons with a Holm correction
 - ▶ Robustness (5-minute variance, subsamples, without 2008 and 2020); a referee-style critique and your reply
- Log each prompt that shaped a decision in `AI_USE.md`; each invented reference, wrong number or look-ahead caught in `AI_ERRORS.md`, with the test that caught it ([templates](#))

What We Learned

- ▣ Parity, bounds and static spanning need no model; the tree and the Black–Scholes equation price by dynamic replication
- ▣ The Greeks measure exposures; discrete hedging leaves an error of order $1/\sqrt{N}$
- ▣ A delta-hedged option is a bet on realised vs implied variance; implied variance exceeded realised variance on average on the S&P 500 and on Bitcoin
- ▣ The surface and the risk-neutral density are estimates: impose no-arbitrage constraints and report bands
- ▣ The VIX is a strip estimator with jump and truncation biases; as a forecast it needs joint tests, logs and HAC loss comparisons

Frontier: Research on Option-Implied Information

- Andersen, Fusari & Todorov (2015): a left-tail factor in index-option surfaces, not spanned by volatility, predicts future equity and variance premia
- Martin (2017): under the negative correlation condition, $R_f \text{SVIX}^2$ bounds the equity premium from below; the bound exceeded 20% at the height of the 2008 crisis
- Bali et al. (2023): nonlinear machine learning on option and stock characteristics predicts equity-option returns out of sample (Chapter 13)
- Open: jointly estimated, arbitrage-free surfaces with valid confidence bands; the pricing of 0DTE options

Coming Next and Further Reading

- Chapter 13: machine learning for financial prediction, with volatility and option-implied features among the inputs
- Options and the implied-volatility surface: Franke, Härdle & Hafner (2019), Härdle, Chen & Overbeck (2017), Čížek, Härdle & Weron (2011), Gatheral (2006)
- Pricing kernels from option prices: Aït-Sahalia & Lo (1998), Grith, Härdle & Park (2013)
- Crypto options and their risk: Matic, Packham & Härdle (2023), Alexander & Imeraj (2020), Pele et al. (2023)

Further Study on Quantinar

Advanced courses on quantinar.com

-  XFG Advanced Methods in Quantitative Finance
-  Pricing Kernels and Risk Premia
-  Hedging Cryptocurrency Options

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_12

References (1/4)

- ▣ Aït-Sahalia, Y., Duarte, J. (2003). Nonparametric option pricing under shape restrictions. *Journal of Econometrics*, 116(1–2), 9–47.
- ▣ Aït-Sahalia, Y., Lo, A.W. (1998). Nonparametric estimation of state-price densities implicit in financial asset prices. *Journal of Finance*, 53(2), 499–547.
- ▣ Alexander, C., Imeraj, A. (2020). The Bitcoin VIX and its variance risk premium. *Journal of Alternative Investments*, 23(4), 84–109.
- ▣ Amihud, Y., Hurvich, C.M. (2004). Predictive regressions: a reduced-bias estimation method. *Journal of Financial and Quantitative Analysis*, 39(4), 813–841.
- ▣ Andersen, T.G., Fusari, N., Todorov, V. (2015). The risk premia embedded in index options. *Journal of Financial Economics*, 117(3), 558–584.
- ▣ Back, K., Crotty, K., Kazempour, S.M. (2022). Validity, tightness, and forecasting power of risk premium bounds. *Journal of Financial Economics*, 144(3), 732–760.
- ▣ Bakshi, G., Kapadia, N. (2003). Delta-hedged gains and the negative market volatility risk premium. *Review of Financial Studies*, 16(2), 527–566.
- ▣ Bakshi, G., Kapadia, N., Madan, D. (2003). Stock return characteristics, skew laws, and the differential pricing of individual equity options. *Review of Financial Studies*, 16(1), 101–143.
- ▣ Bali, T.G., Beckmeyer, H., Mörke, M., Weigert, F. (2023). Option return predictability with machine learning and big data. *Review of Financial Studies*, 36(9), 3548–3602.
- ▣ Bandi, F.M., Fusari, N., Renò, R. (2023). ODTE option pricing. SSRN Working Paper 4503344.
- ▣ Bates, D.S. (1991). The crash of '87: was it expected? The evidence from options markets. *Journal of Finance*, 46(3), 1009–1044.
- ▣ Beckmeyer, H., Branger, N., Gayda, L. (2023). Retail traders love ODTE options... but should they?. SSRN Working Paper 4404704.
- ▣ Bertsimas, D., Kogan, L., Lo, A.W. (2000). When is time continuous?. *Journal of Financial Economics*, 55(2), 173–204.
- ▣ Black, F., Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3), 637–654.
- ▣ Bollerslev, T., Tauchen, G., Zhou, H. (2009). Expected stock returns and variance risk premia. *Review of Financial Studies*, 22(11), 4463–4492.
- ▣ Breeden, D.T., Litzenberger, R.H. (1978). Prices of state-contingent claims implicit in option prices. *Journal of Business*, 51(4), 621–651.
- ▣ Britten-Jones, M., Neuberger, A. (2000). Option prices, implied price processes, and stochastic volatility. *Journal of Finance*, 55(2), 839–866.
- ▣ Brogaard, J., Han, J., Won, P.Y. (2023). Does ODTE options trading increase volatility?. SSRN Working Paper 4426358.
- ▣ Busch, T., Christensen, B.J., Nielsen, M.Ø. (2011). The role of implied volatility in forecasting future realized volatility and jumps in foreign exchange, stock, and bond markets. *Journal of Econometrics*, 160(1), 48–57.
- ▣ Campbell, J.Y., Thompson, S.B. (2008). Predicting excess stock returns out of sample: can anything beat the historical average?. *Review of Financial Studies*, 21(4), 1509–1531.

References (2/4)

- ▣ Carr, P., Wu, L. (2009). [Variance risk premiums](#). *Review of Financial Studies*, 22(3), 1311–1341.
- ▣ Cboe Global Indices (2025). [Cboe Volatility Index: Methodology](#). Cboe Global Markets.
- ▣ Cboe Global Markets. [ODTE options](#). Product page.
- ▣ Chabi-Yo, F., Loudis, J. (2020). [The conditional expected market return](#). *Journal of Financial Economics*, 137(3), 752–786.
- ▣ Christensen, B.J., Prabhala, N.R. (1998). [The relation between implied and realized volatility](#). *Journal of Financial Economics*, 50(2), 125–150.
- ▣ Christie, A.A. (1982). [The stochastic behavior of common stock variances: value, leverage and interest rate effects](#). *Journal of Financial Economics*, 10(4), 407–432.
- ▣ Čížek, P., Härdle, W.K., Weron, R. (Eds.) (2011). [Statistical Tools for Finance and Insurance](#), 2nd ed. Springer.
- ▣ Clark, T.E., West, K.D. (2007). [Approximately normal tests for equal predictive accuracy in nested models](#). *Journal of Econometrics*, 138(1), 291–311.
- ▣ Corsi, F. (2009). [A simple approximate long-memory model of realized volatility](#). *Journal of Financial Econometrics*, 7(2), 174–196.
- ▣ Cox, J.C., Ross, S.A., Rubinstein, M. (1979). [Option pricing: a simplified approach](#). *Journal of Financial Economics*, 7(3), 229–263.
- ▣ Delbaen, F., Schachermayer, W. (1994). [A general version of the fundamental theorem of asset pricing](#). *Mathematische Annalen*, 300(1), 463–520.
- ▣ Demeterfi, K., Derman, E., Kamal, M., Zou, J. (1999). [A guide to volatility and variance swaps](#). *Journal of Derivatives*, 6(4), 9–32.
- ▣ Deribit. [DVOL: Bitcoin volatility index](#). Statistics page.
- ▣ Diebold, F.X., Mariano, R.S. (1995). [Comparing predictive accuracy](#). *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). [Statistics of Financial Markets: An Introduction](#), 5th ed. Springer.
- ▣ Gatheral, J. (2006). [The Volatility Surface: A Practitioner's Guide](#). Wiley.
- ▣ Gatheral, J., Jacquier, A. (2014). [Arbitrage-free SVI volatility surfaces](#). *Quantitative Finance*, 14(1), 59–71.
- ▣ Goyal, A., Welch, I., Zafirov, A. (2024). [A comprehensive 2022 look at the empirical performance of equity premium prediction](#). *Review of Financial Studies*, 37(11), 3490–3557.
- ▣ Grith, M., Härdle, W.K., Park, J. (2013). [Shape invariant modeling of pricing kernels and risk aversion](#). *Journal of Financial Econometrics*, 11(2), 370–399.
- ▣ Hansen, L.P., Hodrick, R.J. (1980). [Forward exchange rates as optimal predictors of future spot rates: an econometric analysis](#). *Journal of Political Economy*, 88(5), 829–853.

References (3/4)

- Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). *Applied Quantitative Finance*, 3rd ed. Springer.
- Harrison, J.M., Pliska, S.R. (1981). Martingales and stochastic integrals in the theory of continuous trading. *Stochastic Processes and their Applications*, 11(3), 215–260.
- Heston, S.L. (1993). A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Review of Financial Studies*, 6(2), 327–343.
- Hodrick, R.J. (1992). Dividend yields and expected stock returns: alternative procedures for inference and measurement. *Review of Financial Studies*, 5(3), 357–386.
- Hull, J., White, A. (1987). The pricing of options on assets with stochastic volatilities. *Journal of Finance*, 42(2), 281–300.
- Jackwerth, J.C. (2000). Recovering risk aversion from option prices and realized returns. *Review of Financial Studies*, 13(2), 433–451.
- Jackwerth, J.C., Rubinstein, M. (1996). Recovering probability distributions from option prices. *Journal of Finance*, 51(5), 1611–1631.
- Jiang, G.J., Tian, Y.S. (2005). The model-free implied volatility and its information content. *Review of Financial Studies*, 18(4), 1305–1342.
- Kodde, D.A., Palm, F.C. (1986). Wald criteria for jointly testing equality and inequality restrictions. *Econometrica*, 54(5), 1243–1248.
- Lee, R.W. (2004). The moment formula for implied volatility at extreme strikes. *Mathematical Finance*, 14(3), 469–480.
- MacKenzie, D., Millo, Y. (2003). Constructing a market, performing theory: the historical sociology of a financial derivatives exchange. *American Journal of Sociology*, 109(1), 107–145.
- Martin, I. (2017). What is the expected return on the market?. *Quarterly Journal of Economics*, 132(1), 367–433.
- Matic, J.L., Packham, N., Härdle, W.K. (2023). Hedging cryptocurrency options. *Review of Derivatives Research*, 26(1), 91–133.
- Merton, R.C. (1973). Theory of rational option pricing. *Bell Journal of Economics and Management Science*, 4(1), 141–183.
- Merton, R.C. (1976). Option pricing when underlying stock returns are discontinuous. *Journal of Financial Economics*, 3(1–2), 125–144.
- Mincer, J., Zarnowitz, V. (1969). The evaluation of economic forecasts. In *Economic Forecasts and Expectations*, NBER, 3–46.
- Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.
- Pele, D.T., Mazurencu-Marinescu, M. (2012). Modelling stock market crashes: the case of Bucharest Stock Exchange. *Procedia – Social and Behavioral Sciences*, 58, 533–542.
- Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *European Journal of Finance*, 29(10), 1064–1105.

References (4/4)

- ▣ Rubinstein, M. (1994). Implied binomial trees. *Journal of Finance*, 49(3), 771–818.
- ▣ Stambaugh, R.F. (1999). Predictive regressions. *Journal of Financial Economics*, 54(3), 375–421.
- ▣ Welch, I., Goyal, A. (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4), 1455–1508.
- ▣ Whaley, R.E. (2000). The investor fear gauge. *Journal of Portfolio Management*, 26(3), 12–17.
- ▣ Zulfikar, N., Gulzar, S. (2021). Implied volatility estimation of bitcoin options and the stylized facts of option pricing. *Financial Innovation*, 7, 67.